

Part - A

1. What is inferential statistics?

Inferential statistics is used to study sample and make conclusions about a larger population.

It helps us to make predictions about the population using sample data.

It is useful when there is no possibility to collect data from every person or item of the population.

2. What is hypothesis testing and its components?

Hypothesis testing is used to check whether there is enough evidence to support a statement about a population.

It helps us to decide whether a particular claim should be accepted or rejected based on sample.

• Null Hypothesis (H_0):

basic assumption that there is no effect, or no relation between variables. Tested using sample data.

• Alternative Hypothesis (H_1):

there is an effect, difference, or relation between variables. It is accepted when there is enough evidence against H_0 .

3. Explain confidence interval and critical value.

Confidence Interval is a range of values which is used to estimate the true values of a population. For example, a 95% confidence interval gives a range in which the population value is expected to lie.

A confidence interval also shows the possible range of the actual population value.

Critical Value is a value to decide whether we should reject the null hypothesis or not.

It depends on the significance level (α).

If test value goes beyond the critical value, H_0 is rejected.

4. Define P-Value.

A P-Value tells us how likely the observed result is if null hypothesis is true.

A small P-Value gives strong evidence against H_0 . Usually, if the p-value is less than or equal to 0.05, we reject H_0 . If P-Value is greater than 0.05, we normally do not reject H_0 .

The P-value is useful for making a decision in hypothesis testing.

5. Differentiate Type I and Type II errors.

Type I Error :

Happens when H_0 is true, but we reject it. also called False Positive.

In simple words, there is an effect when there is actually no effect.

Type II Error :

Happens when H_0 is false, but we don't reject it. also called False Negative.

In simple words, we failed to identify effect that actually exist.

6. Brief descriptions of z-test, t-test, chi-square test, and ANOVA test.

Z-test is used to compare population means when the sample size is large ($n \geq 30$) and the population standard deviation is known and population variance is known.

T-test is used to compare the means of one or two groups when the sample size is small ($n < 30$) or the population variance is known. It is commonly used when the sample standard deviation is used.

Chi-Square Test is used to check the relation or independence between categorical deviation variables. It can also be used to compare observed and expected frequencies.

ANOVA is used to compare the means of three or more groups. It helps to check whether there is a significant difference between the group means. ANOVA is useful when we want to compare more than two groups at the same time.

7. What is Covariance?

Covariance shows the direction of the relation between two variables. It tells us whether two variables tend to change in the same direction or in opposite directions.

- Positive covariance: Both variables tend to increase or decrease together.
- Negative covariance: When one variable increases, the other tends to decrease.
- Zero or near zero: It shows little or no linear movement between two variables.

Covariance mainly tells us the direction of the relation, not its exact strength.

8. Correlation: shows the strength and direction of relation between two variables.

value is between -1 and $+1$

near

- $+1$ shows strong positive relation
- near -1 shows strong negative relation
- near 0 shows weak or no linear relation